

Background for data imputation

- In clinical trials, we collect data at predetermined timepoints
- These predetermined timepoints are called visits
- Sometimes, the data could not be collected/reported at some visits for reasons like subject refused for a test or collected sample damaged etc.
- For analysis purposes, we may want to have all data available for each visit for all subjects
- We fill in the missing visits with the results from a prespecified algorithm
- This concept of 'filling' in missing data is called as data imputation

Some common approaches of data imputation

- LOCF - Last observation carried forward
-  LOCF - ~~Baseline~~ observation carried forward
- BOCF - Best observation carried forward
- WOCF - Worst observation carried forward
- INTERPOLATION - Average result of previous and next non-missing observations is populated

How do we represent imputed records in ADaM

- CDISC ADaM standard allows for data imputation
- As per the standard, we can create new derived records within the parameter
- These new records have to be differentiated from the source records by populating the DTYPE variable